Luiss
Libera Università Internazionale
degli Studi Sociali Guido Carli

Course overview

Introduction to Computer Programming

Management and Artificial Intelligence

LUISS

Greetings, folks!

Welcome to the ***Introduction to Computer Programming*** course!

Teachers

Emilio Coppa

- Position: Associate Professor @ LUISS
- Research topics: software testing, vulnerability detection
- Other info: https://ecoppa.github.io
- Office Hours: send me an e-mail so we can arrange a time slot.
- Contact: `ecoppa@luiss.it`

Teaching assistants

Susanna Caroppo

- Position: Postdoc at Roma Tre University
- Research topics: quantum algorithms, graph theory, and combinatorial optimization
- Other info: https://compunet.ing.uniroma3.it/#!/people/caroppo
- Office Hours: send me an e-mail so we can arrange a time slot.
- Contact: `scaroppo@luiss.it`

Mattia Cervellini

- PhD Student in ICT @ Sapienza
- LUISS graduate in Management and Artificial Intelligence (formerly Management and Computer Science)
- Research topics: AI for complex systems, bioinformatics, (hyper)graphs, representation learning
- Office Hours: drop me an e-mail so we can arrange a time slot.
- Contact: `mcervellini@luiss.it`

Important Resources

Web Resources you should have access to:

- **MyLUISS**
- **Course Syllabus**
- **Rooms and Timing**

e-Learning platform

MyLUISS is our e-Learning platform: ***It is important that you get access!***

You'll find there:

- **Slides** covering:
  - theory
  - exercises
- **Details** on exam dates, reservations, results (besides standard Luiss channels)
- **Communications** from teacher and teaching assistants

A live and editable version of the course material is available at **https://introcp.github.io/**

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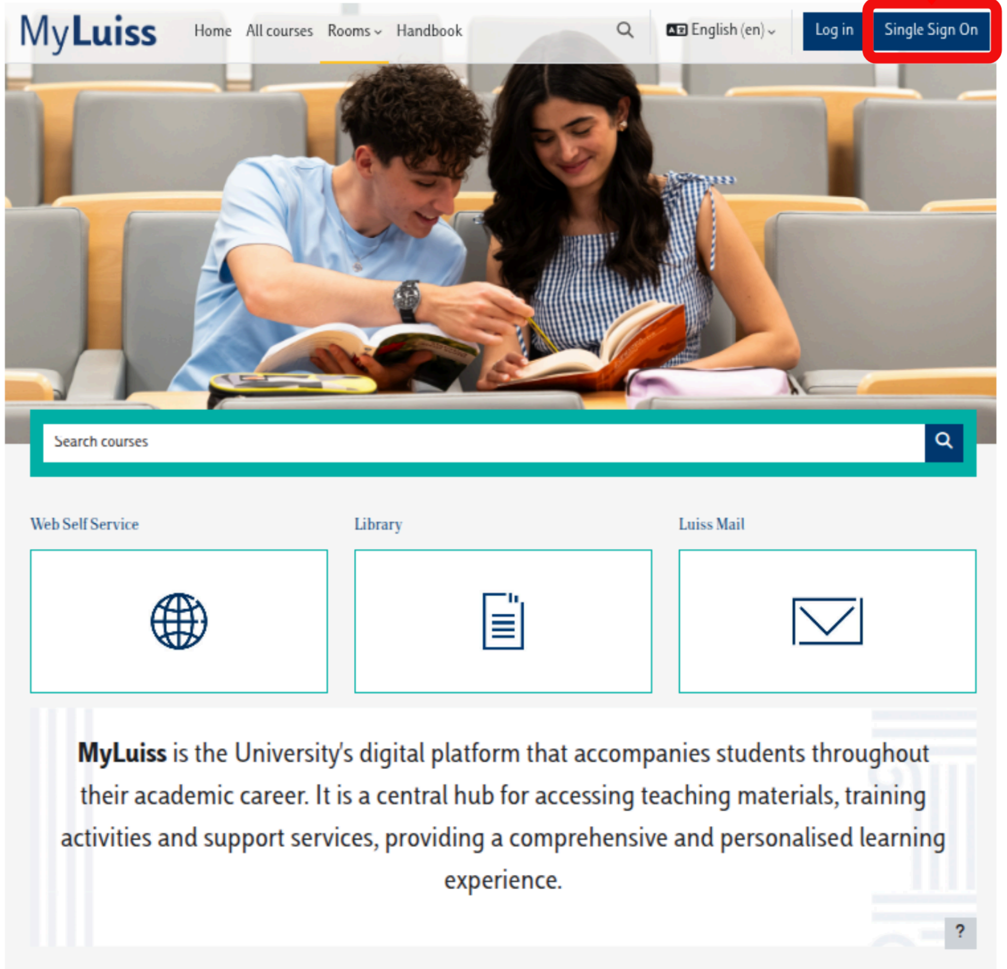
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MyLuiss is the University's digital platform that accompanies students throughout their academic career. It is a central hub for accessing teaching materials, training activities and support services, providing a comprehensive and personalised learning experience.
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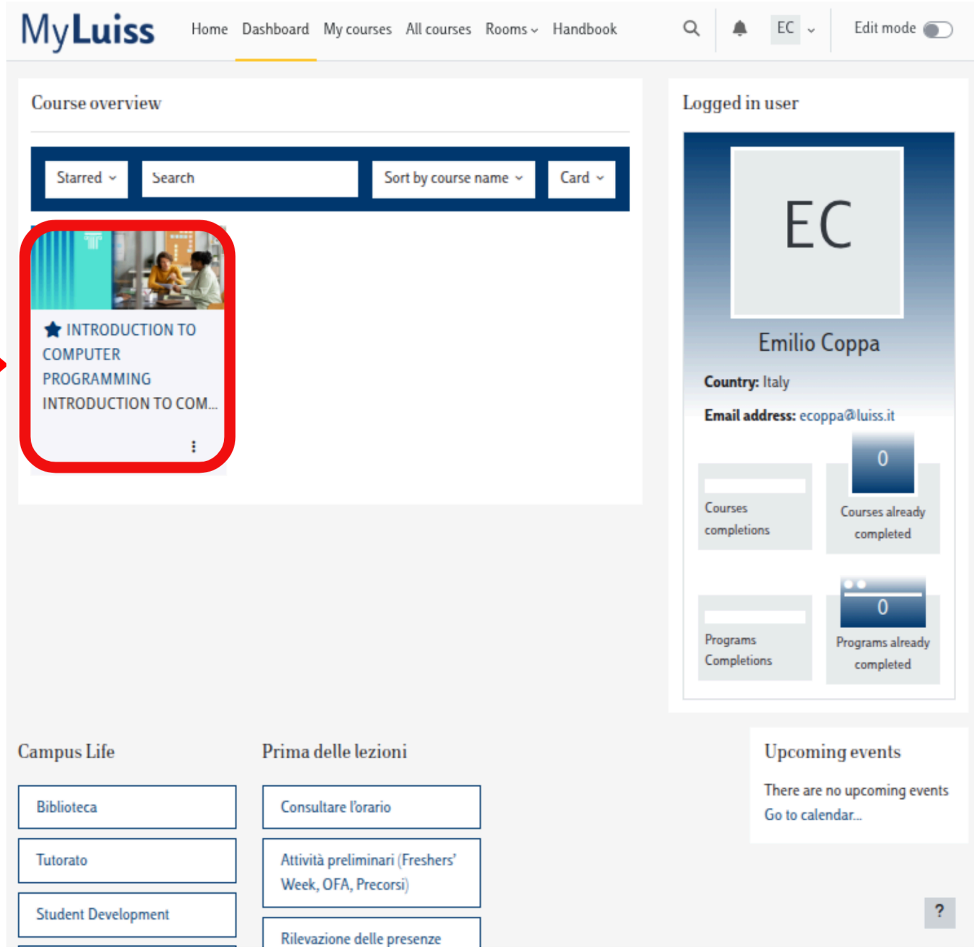
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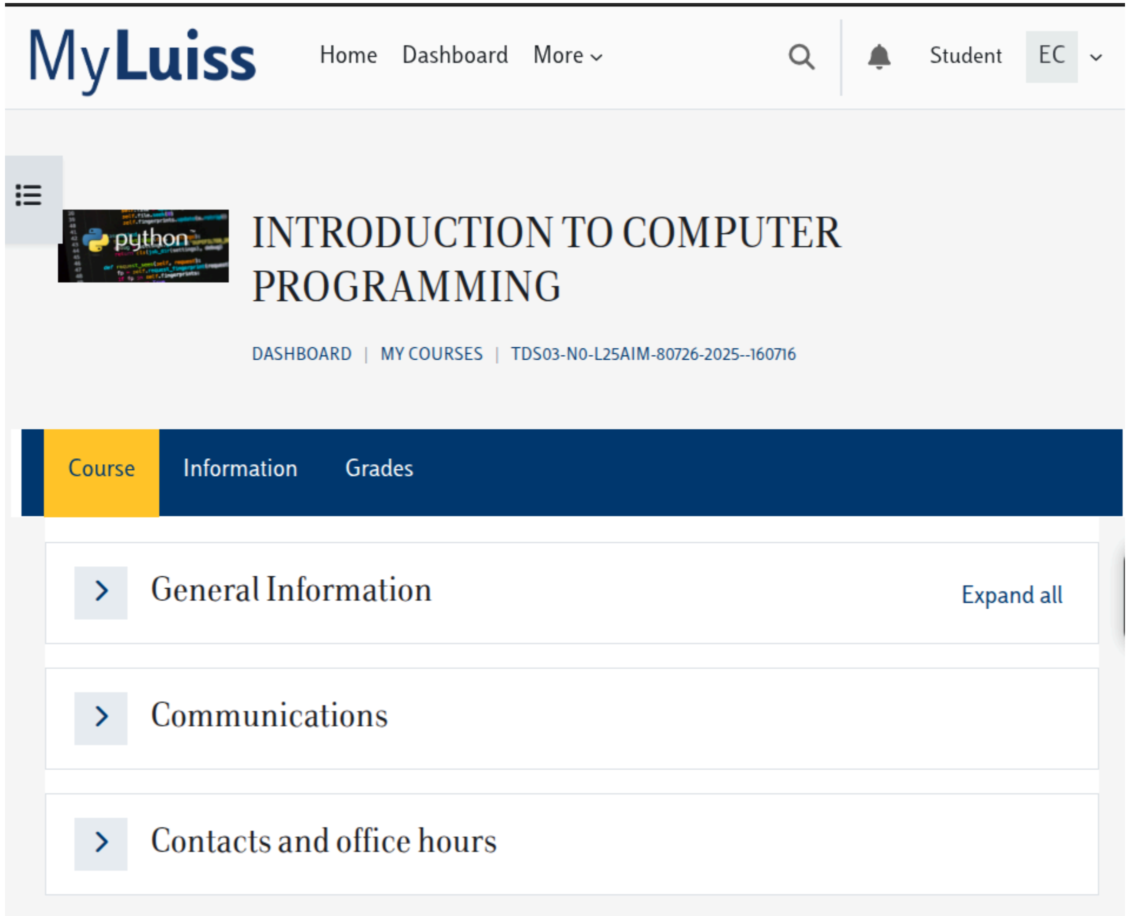
Prima delle lezioni
Consultare l'orario
Attività preliminari (Freshers' Week, OFA, Precorsi)
Rilevazione delle presenze
Upcoming events
There are no upcoming events
Go to calendar...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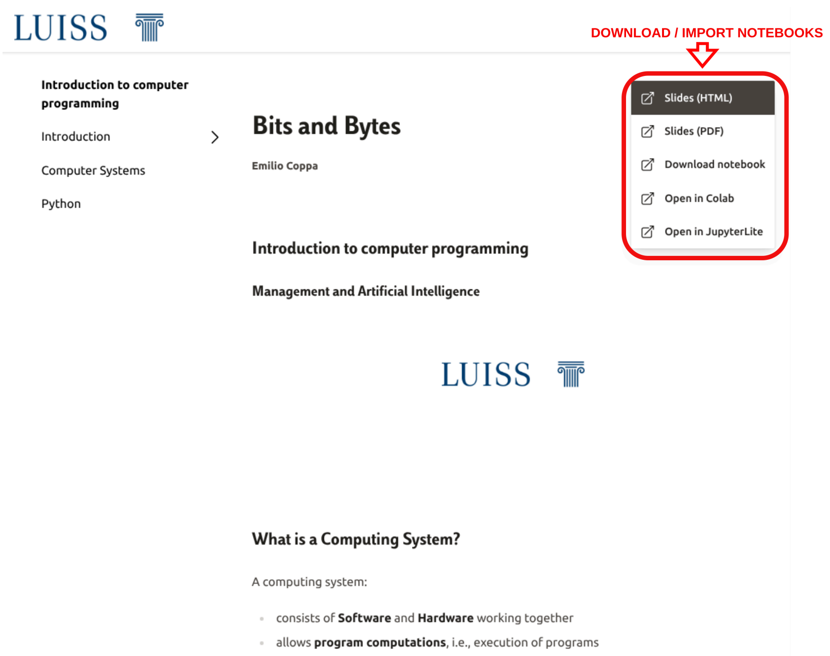
Access the *live* material

Material will be made available on MyLUISS via slides. However, you can have access to:

- browsable content of the course
- editable notebooks with theory materials
- notebooks with exercises (easy to download or import on Google Colab and Jupyter Lite)

by accessing https://introcp.github.io/



Course Goals

- The course is intended to teach **basic programming concepts** to students with no prior coding experience, developing an attitude towards **computational thinking**.

- It will provide the students with:
    - an understanding of the role that computation can play in solving problems
    - the ability to write small programs to accomplish useful goals

Course Roadmap

The course will familiarize the students with computer programming and will cover the following topics:

- Binary numbers and boolean logic
- Computer hardware
- Computer software
- Operating Systems essentials
- Introduction to programming languages
- Python basics
- Variables, data types, expressions
- Branching
- Functions and modules
- Recursion
- Strings and Lists
- Dictionaries, Tuples, and Sets
- Object-oriented programming and classes

Organization

Lectures + **Lab Sessions**: *Laptops at your fingertips, always!*

- **Monday** @ 13:45-15:00 (75 minutes) (often) Theory lectures & some exercises
- **Tuesday** @ 11:30-13:00 (90 minutes) (often) Theory lectures & some exercises
- **Wednesday** @ 10:00-11:30 (90 minutes) (often) Lab Session with TA
- **Friday** @ 17:30 - 18:45 (75 minutes) (often) Lab Session with TA

Exam

Written exam (2 hours):

- **Coding** (50% of the overall grade): 5 coding exercises
- **Theory** (50% of the overall grade): 15 multiple-choice quizzes

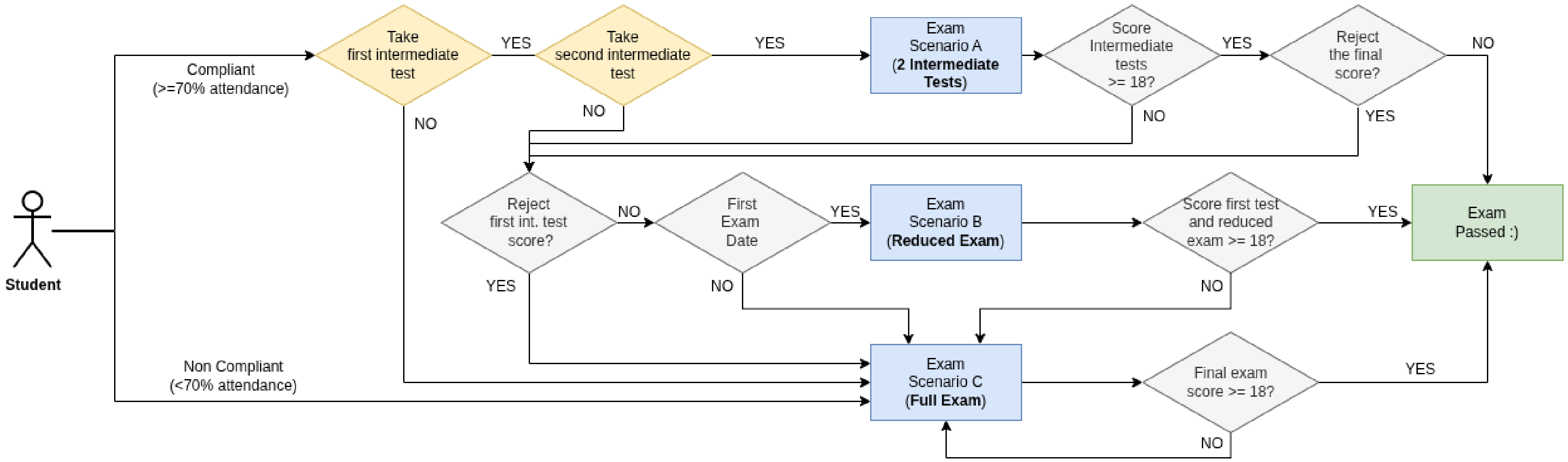
In light of the LUISS Continuous Assessment verification method, we will be having 2 *in itinere* exams:

- *First Intermediate Test* (week 7: 19-23 October)
    - 60 minutes, 8 theory quizzes, 2 coding exercises
- *Second Intermediate Test* (week 13: 30 November - 4 December)
    - 75 minutes, 7 theory quizzes, 3 coding exercises

Each intermediate test is a short exam on the topics covered so far. It will consist of a mixture of coding exercises and theory questions (multiple-choice quiz). Note that:

- the sum of the coding parts between the two tests yields 31 points
- the sum of the theory parts between the two tests yields 31 points

NOTE: you must be a ***compliant student*** (at least 70% of attendance) to take the intermediate tests.



Student
Compliant
(>=70% attendance)
Non Compliant
(<70% attendance)
Take
first intermediate
test
YES
NO
Take
second intermediate
test
YES
NO
Exam
Scenario A
(**2 Intermediate Tests**)
Score
Intermediate
tests
>= 18?
YES
NO
Reject
the final
score?
NO
YES
Reject
first int. test
score?
NO
YES
First
Exam
Date
YES
NO
Exam
Scenario B
(**Reduced Exam**)
Score first test
and reduced
exam >= 18?
YES
NO
Exam
Passed :)
Exam
Scenario C
(**Full Exam**)
Final exam
score >= 18?
YES
NO

Exam scenario A: compliant student passing both tests

A compliant student taking the two intermediate tests has passed the exam if:

- the sum of the coding parts from the two tests is greater than 18
- the sum of the theory parts from the two tests is greater than 18

The final exam score is computed as the average of two scores.

The student can:

- accept the score:
  - GREAT!
  - **the student has to book in the first exam date**
  - there is no need to take the exam during the exam date!
- reject the score of the second test: see *Exam scenario B*
- reject the score at both tests: see *Exam scenario C*

Exam scenario B: reduced exam

A compliant student failing (too low score) or missing the second test can take a **reduced** exam (which is similar to the second intermediate test: 75 minutes, 7 theory quizes, 3 coding exercises) **during the first exam date**.

The student passes the exam if:

- the sum of the coding parts from the first test and the reduced exam is greater than 18
- the sum of the theory parts from the first test and the reduced exam is greater than 18

The final exam score is computed as the average of these two scores.

The student cannot reject the final exam score if it is equal to or greater than 18.

Students failing the exam: see *Exam scenario C*

Exam scenario C: other situations

Students that:

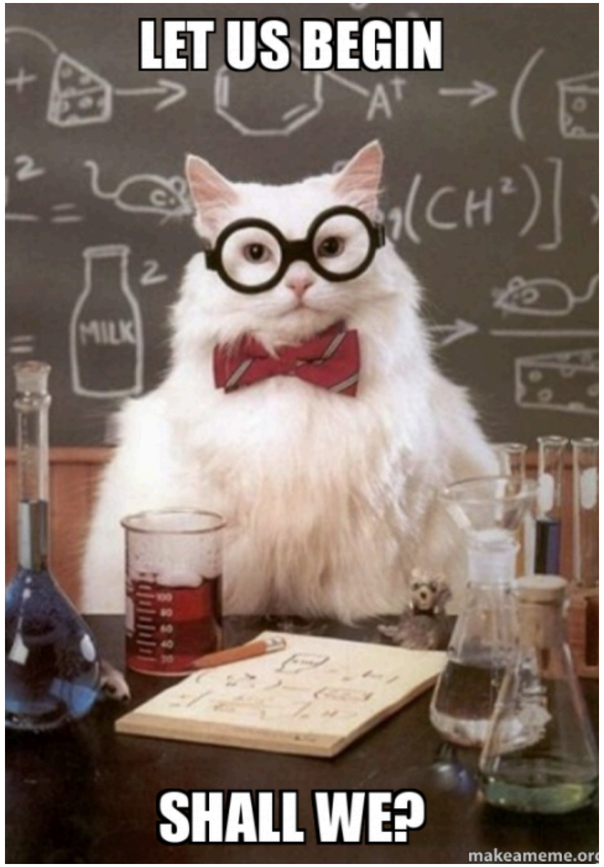
- **are not compliant**
- **are taking the exam after the first exam date**
- are compliant but missed both tests
- have taken the tests but failed both tests
- have rejected the score of both tests
- have attempted the reduce exam (scenario B) and failed it

have to take the **full** exam (120 minutes):

- 15 theory quizes (20 minutes)
- 5 coding exercises (100 minutes)

The final exam score is computed as the average of two parts (theory and coding).

Students cannot reject the score if it is equal to or greater than 18.


LET US BEGIN
SHALL WE?